

ECON 1550 International Finance

Output and the Exchange Rate in the Short Run

Behavioral equations in the goods market

$$Y = C + I + G + CA$$

$$\begin{array}{l} \text{DEMAND FOR} \\ \text{DOMESTIC} \\ \text{GOOD} \end{array} = C + I + G + CA$$

$$\begin{array}{l} \text{SUPPLY FOR} \\ \text{DOMESTIC} \\ \text{GOOD} \end{array} = Y$$

Demand for domestic goods

$$\begin{array}{l} \text{DEMAND} \\ \text{FOR DOM} \\ \text{GOODS} \end{array} = C + I + G + CA = D$$

$$\begin{array}{l} \text{DOMESTIC} \\ \text{DEMAND} \\ \text{FOR GOODS} \end{array} \equiv A = C + I + G$$

$$A + EX - IM = A + CA = D$$

Demand for domestic goods

$$C = C(Y_D^{(+1)}) \quad \text{CONSUMPTION (OF DOM. AND FOREIGN)}$$

$$Y_D \equiv Y - T \quad \text{DISPOSABLE INCOME}$$

$$I = \bar{I} \quad \text{EXOGENOUS INVESTMENT}$$

$$G = \bar{G}$$

"

GOV. PURCHASES

} BOTH DOM
AND FOR.
GOODS

$$CA = CA(q, Y - T, Y^*)$$

$$q \equiv \frac{EP^*}{P} = \frac{\text{DOLLAR PRICE OF FOREIGN GOOD}}{\text{DOLLAR PRICE OF DOM GOOD}}$$

Demand for domestic goods

$$CA = EX - IM$$

$$EX = EX(q, Y^*)$$

(+1) (+1)

$$IM = IM(q, Y_D)$$

(-) (+1)

$q \uparrow$ VOLUME & PRICE $q \uparrow$
VOLUME DOMINATES

$q \uparrow$ REAL DEPR.

$$\frac{EP^*}{P} = q$$

$\Rightarrow$ HOME
 GOOD
 APPEARS
 CHEAPER

$$\underbrace{IM}_{\text{DOM GOOD}} = \underbrace{\frac{\text{PRICE}}{\text{FOREIGN GOOD}}}_{\equiv q} \times \underbrace{VOLUME}_{\text{FOREIGN GOOD}}$$

A short-run model of the goods market

Exogenous variables

Variable	Description
E	Nominal exchange rate
I	Investment
G	Government spending
T	Taxes
P	Price level
P^*	Foreign price level
Y^*	Foreign income

Endogenous variables

Variable	Description	Equation	Type of equation
Y	Income, production	$Y = D$	Equilibrium condition
Y_D	Disposable income	$Y_D \equiv Y - T$	Identity
EX	Exports	$EX = EX(\underset{(+)}{q}, \underset{(+)}{Y^*})$	Behavioral
IM	Imports	$IM = IM(\underset{(-)}{q}, \underset{(+)}{Y_D})$	Behavioral
CA	Current account	$CA \equiv EX - IM = CA(\underset{(+)}{q}, \underset{(-)}{Y_D}, \underset{(+)}{Y^*})$	Identity
D	Demand for domestic goods	$D \equiv C + I + G + CA$	Identity
A	Domestic demand	$A \equiv C + I + G$	Identity
C	Consumption	$C = C(\underset{(+)}{Y_D})$	Behavioral
q	Real exchange rate	$q \equiv \frac{EP^*}{P}$	Identity

DD Curve

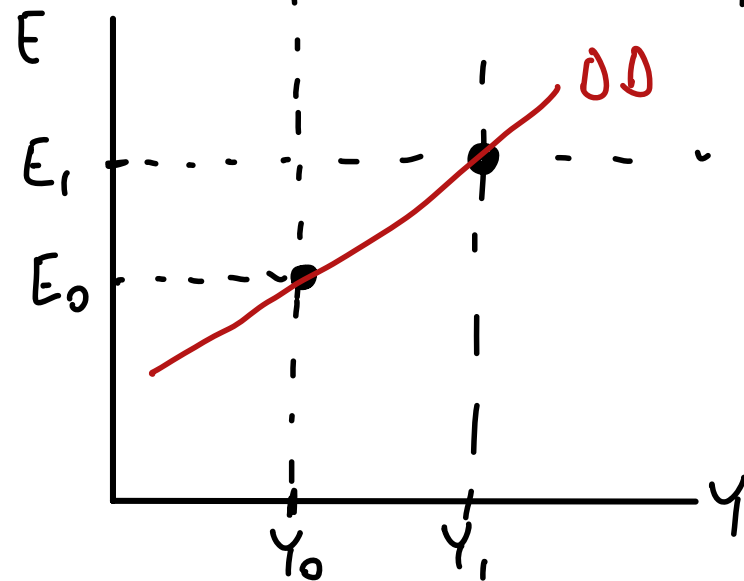
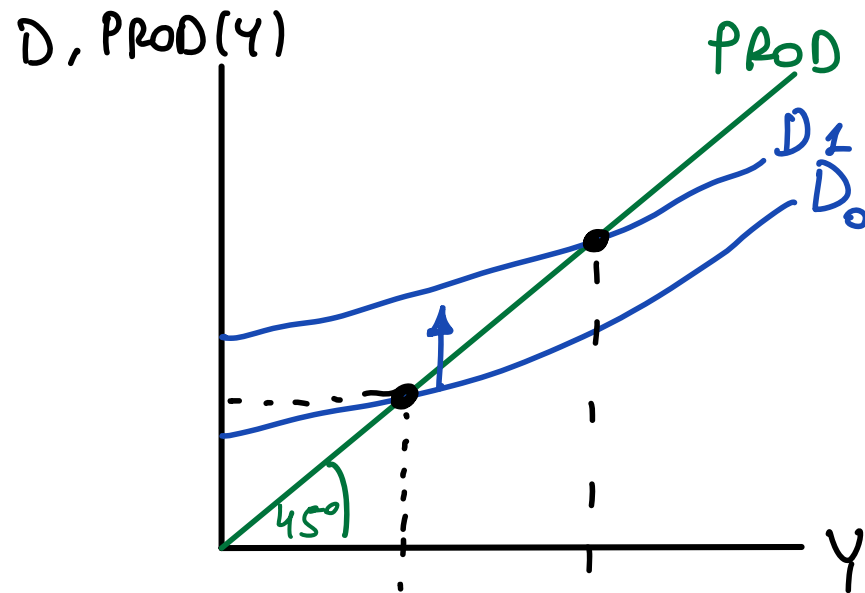
$$Y = D$$

$$Y = C + I + G + CA$$

$$Y = C_{(+)}(Y_D) + I + G + CA_{(+)}(q_{(+)} Y_{D(-)}, Y^*_{(+)})$$

$$Y = C_{(+)}(Y_D) + I + G + CA_{(+)}(EP^*/P_{(+)} Y_{D(-)}, Y^*_{(+)})$$

$Y \uparrow \rightarrow C \uparrow$ BUT $CA \downarrow$. HOWEVER,
 D ALWAYS $\uparrow$



$$D = C + I + G + NX \quad \text{WHEN } Y \uparrow$$

$C \uparrow$ $NX \downarrow$ SO D ?

NOT UNCLEAR, $D \uparrow$

C : CONS. OF DOM. AND FOREIGN GOODS

$C - IM$ = DOM DEM. FOR DOM GOOD

$$NX = EX - IM$$

EX DOES NOT
CHANGE

IM GOES
UP

$$C = C_{\text{DOM GOOD}} + C_{\text{FOREIGN GOOD}}$$

IM = DOM DEMAND FOR
FOREIGN

$$D = C + I + G + NX - IM$$

$$= C - IM + \dots$$

$$= C_{\text{DOM GOOD}}$$

$$C = C(Y) = C_{\text{DOM}}(Y) + C_{\text{FOREIGN}}(Y)$$

$$IM = IM(Y)$$

$$C - IM = C(Y) - IM(Y) =$$

$$= C_{\text{DOM}}(Y) + \cancel{C_{\text{FOREIGN}}(Y)} - \cancel{IM(Y)}$$

$$= C_{\text{DOM}}(Y)$$

Short-run FX and money market model

Exogenous variables

Variable	Description
R^*	Foreign interest rate
E^e	Expected exchange rate
Y	Real income
M^s	Money supply
P	Price level

Endogenous variables

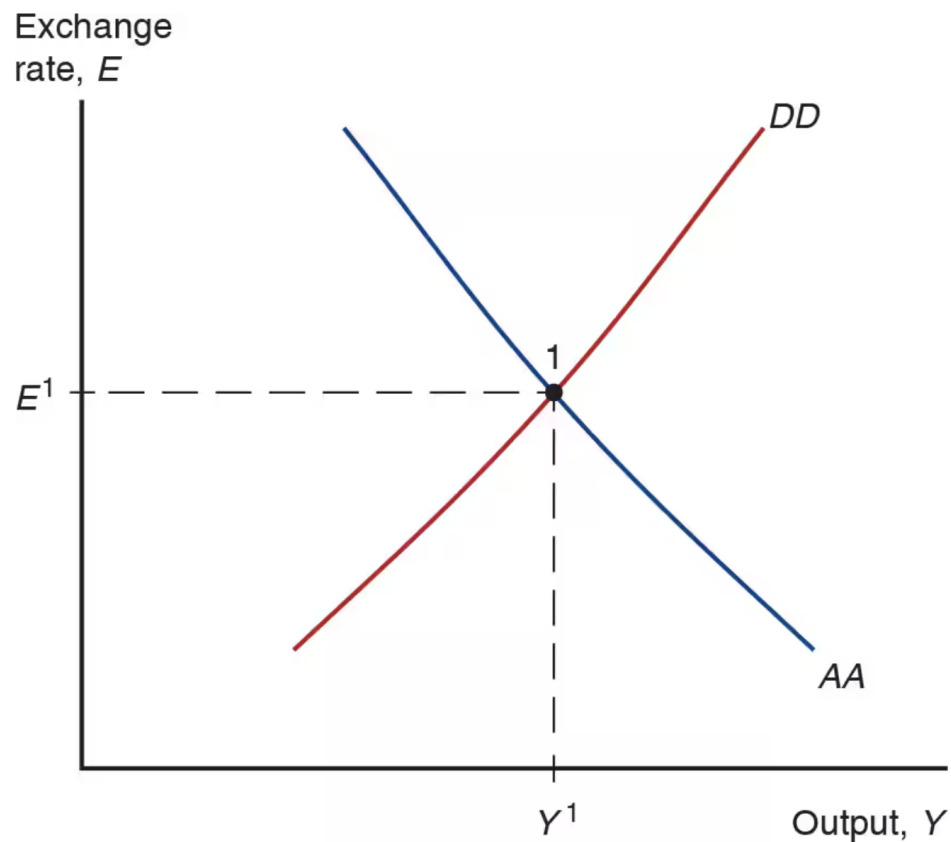
Variable	Description	Equation	Type of equation
E	Exchange rate	$R = R^* + \frac{E^e - E}{E}$	Equilibrium condition
R	Domestic interest rate	$M^d/P = L(R, Y)$	Behavioral equation
M^d	Money demand	$M^d = M^s$	Equilibrium condition

AA Curve

$$\text{(UIP):} \quad R = R^* + \frac{E^e}{E} - 1$$

$$\text{(MS) = (MD):} \quad \frac{M^s}{P} = L(R, Y)$$

Short-run Equilibrium in *AA-DD* model



Example: DD Schedule

$$C_{(+)}(Y_D) = 1 + 0.75Y_D$$

How to find the DD schedule

$$EX_{(+)}(q_{(+)}, Y^*) = 0.15 + 0.5q + 0.1Y^*$$

$$IM_{(-)}(q_{(-)}, Y_D) = 0.1 - 0.2q + 0.12Y_D$$

$$Y_D = Y - T$$

$$q = \frac{EP^*}{P}$$

Example: AA Schedule

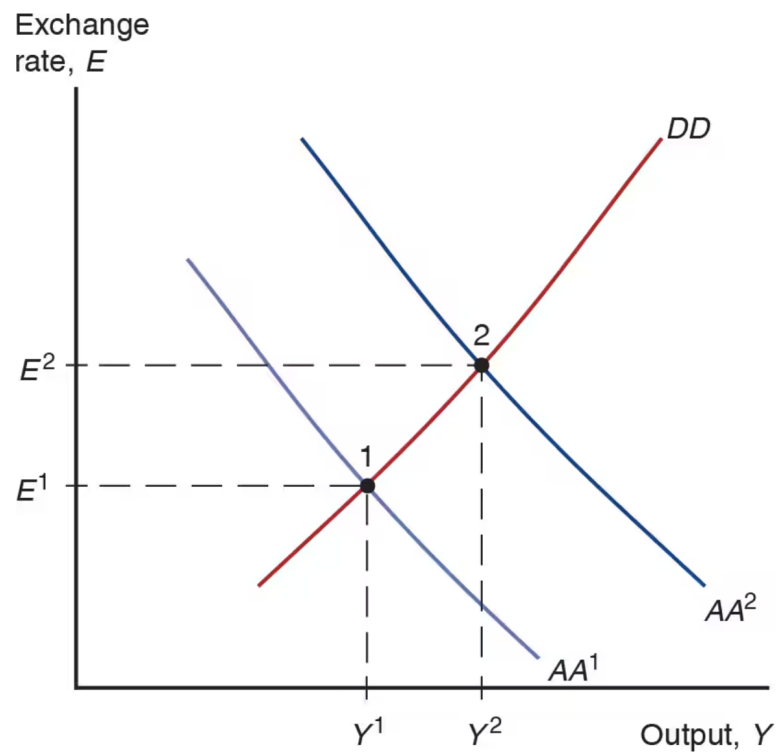
$$R = R^* + \frac{E^e - E}{E}$$

How to find the AA schedule

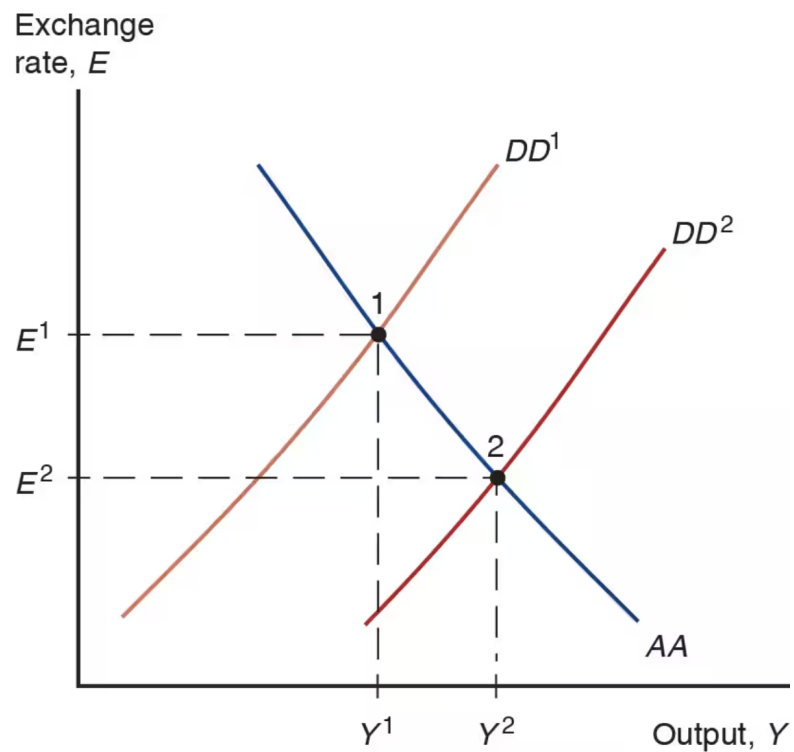
$$\frac{M^d}{P} = \frac{M^s}{P}$$

$$\begin{aligned}\frac{M^d}{P} &= L\left(R, Y\right)_{(-) (+)} \\ &= 1.1 - R + 0.05Y\end{aligned}$$

Temporary Change in Monetary Policy



Temporary Change in Fiscal Policy



Full AA-DD Model

DD Schedule: $Y = C(Y_D) + I + G + CA(EP^*/P, Y_D, Y^*)$

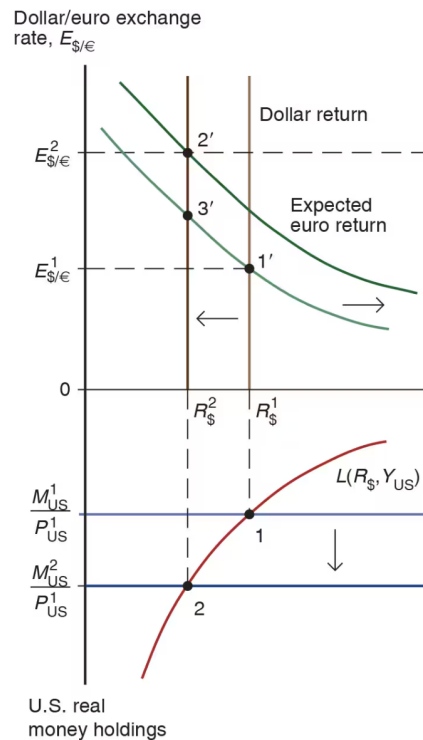
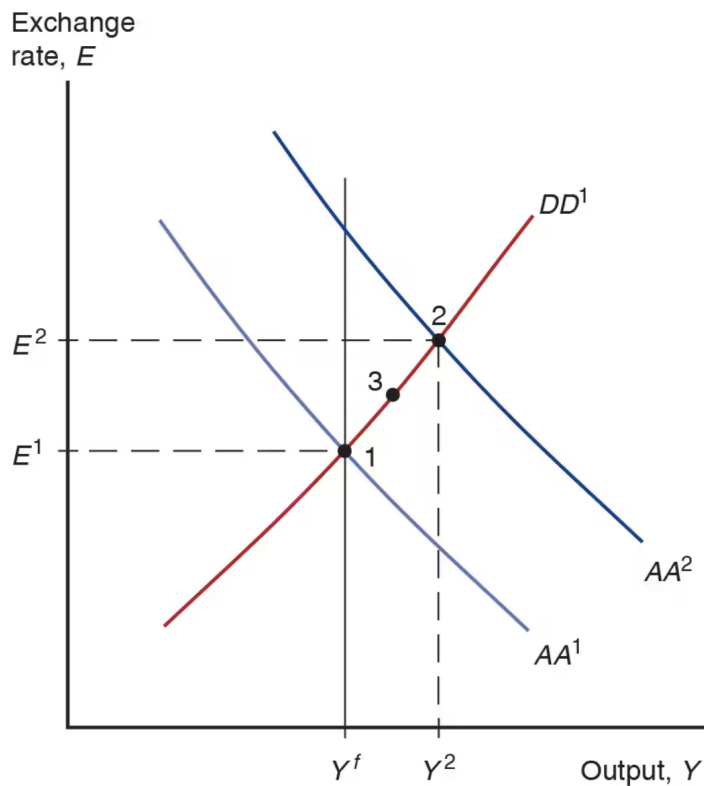
AA Schedule: $\frac{M^s}{P} = L\left(R^* + \frac{E^e}{E} - 1, Y\right)$

Phillips Curve: $\pi = \pi^e + \alpha(Y - Y^f)$

Definition of inflation: $\pi_t = \frac{P_t}{P_{t-1}} - 1$

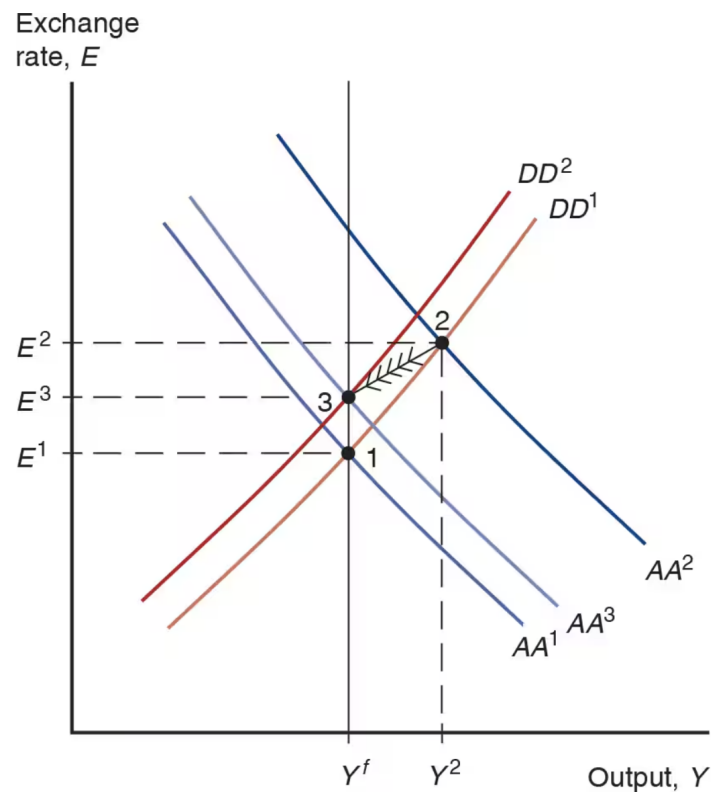
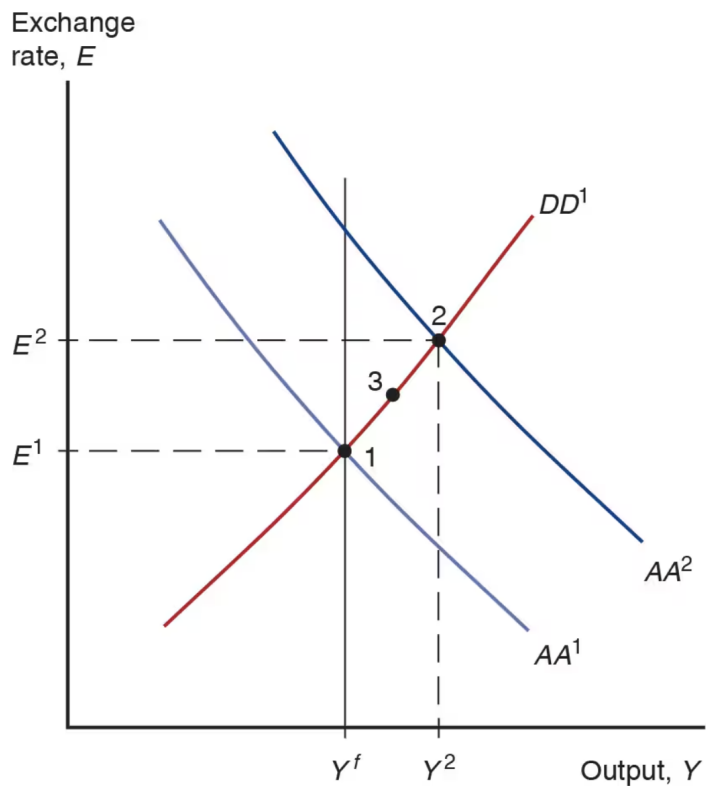
Definition of expected inflation: $\pi_t^e = \frac{P_{t+1}^e}{P_t} - 1$

Permanent Shifts in Monetary Policy

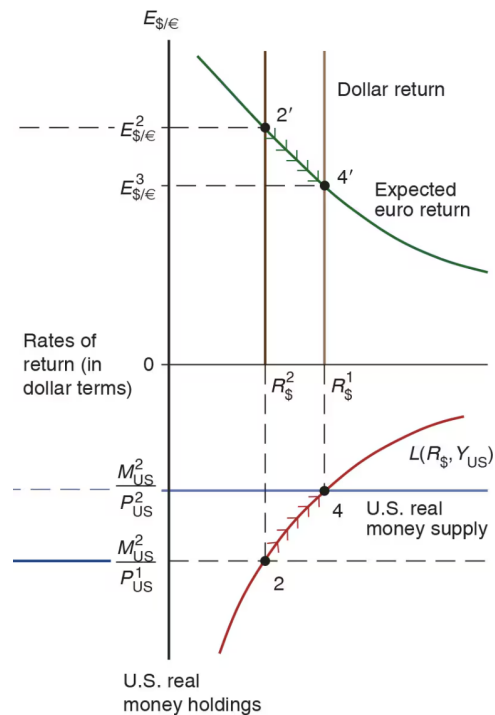
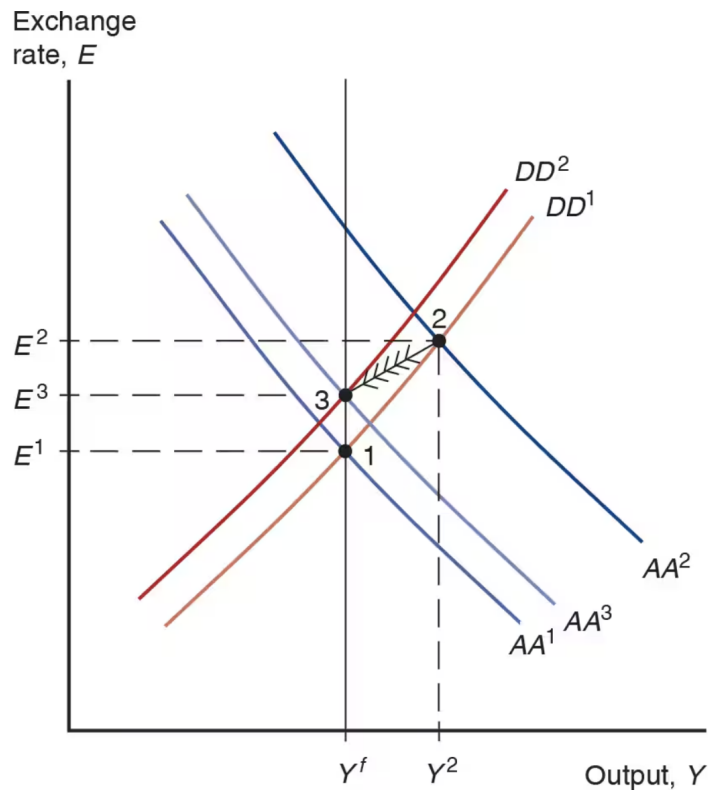


(a) Short-run effects

Permanent Shifts in Monetary Policy

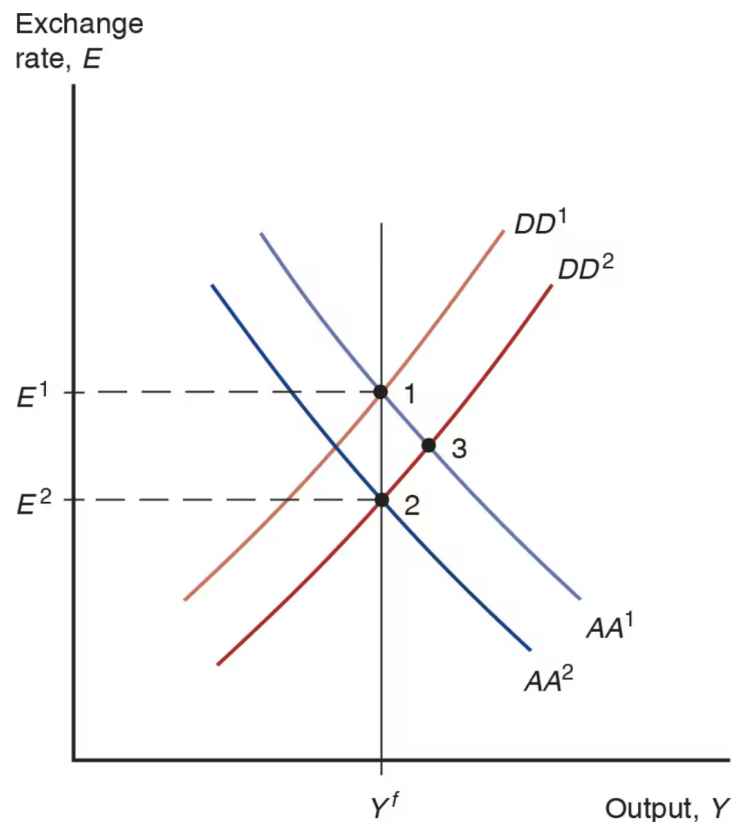


Permanent Shifts in Monetary Policy



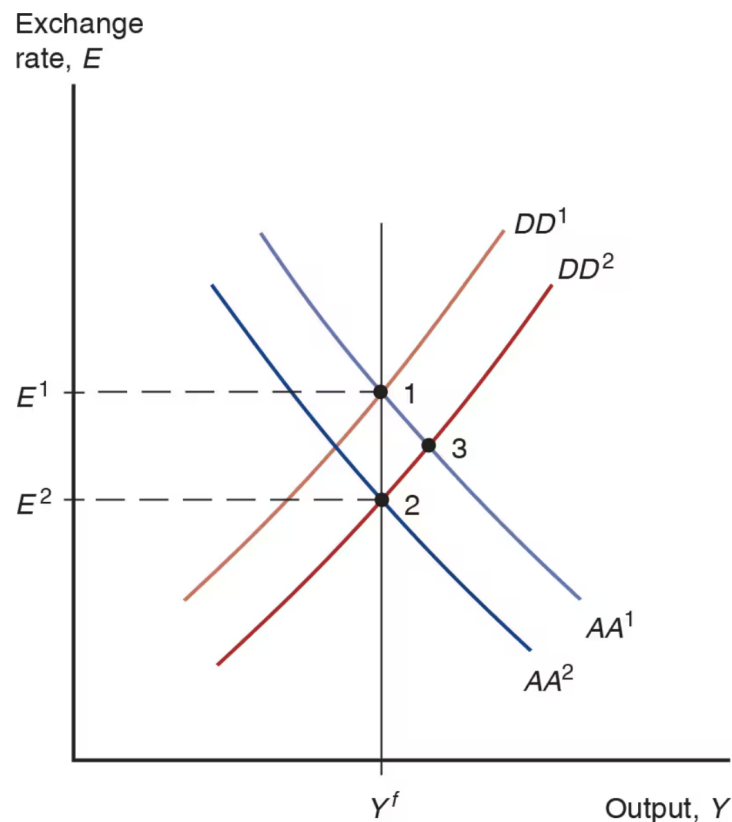
(b) Adjustment to long-run equilibrium

Permanent Shifts in Fiscal Policy



- Start in a medium run equilibrium at point 1 with $R_0 = R^*$ and $E_0^e = E^1$
- Increase in G shifts DD to the right
- Since the shift in DD is permanent,
- At point 3, exchange rate is lower than E_1
- appreciation at point 3 is also expected to be permanent at point 3
- E^e must also go down
- Lower E^e shifts AA down
- But how much?

Permanent Shifts in Fiscal Policy



- There are no more shifts expected to occur, E^e remains constant
- Therefore $E = E^e$
- Real money supply is constant
- Real money demand must stay unchanged
- UIP implies R does not change
- Y must be unchanged as well to keep real money demand unchanged
- If Y must be equal to its original level Y^f , AA^1 must shift to AA^2

DD Schedule with Tariffs

Slope of DD Curve with Tariffs